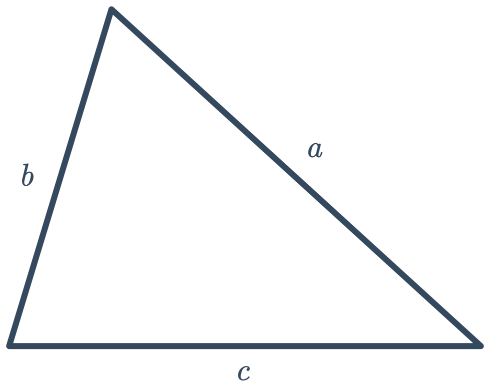
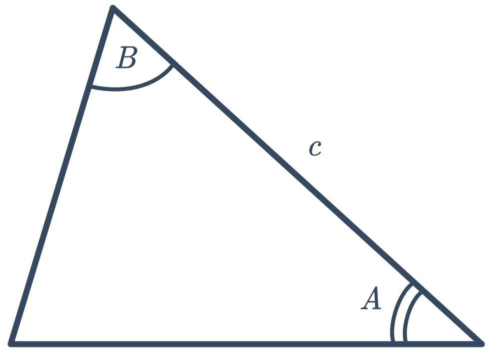
Sine rule

1 For each of the given triangles, determine if there is enough information to find all the remaining sides and angles in the triangle using only the sine rule:

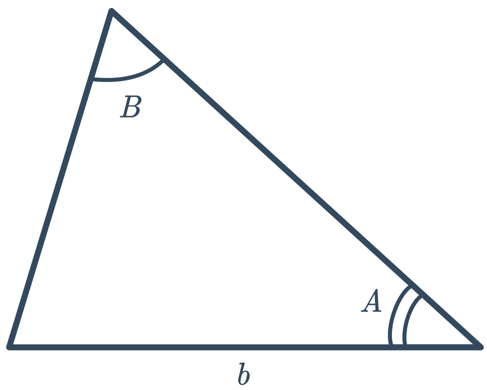
a Three sides are known:



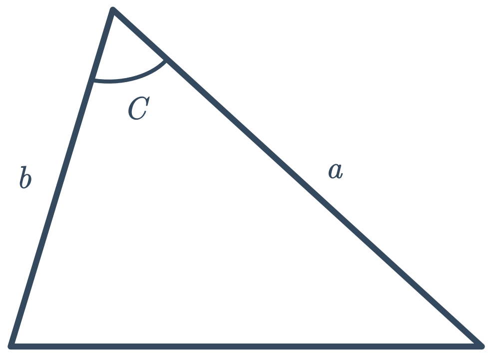
b Two of the angles and the side included between them are known:



c Two of the angles and a side not included between them are known:

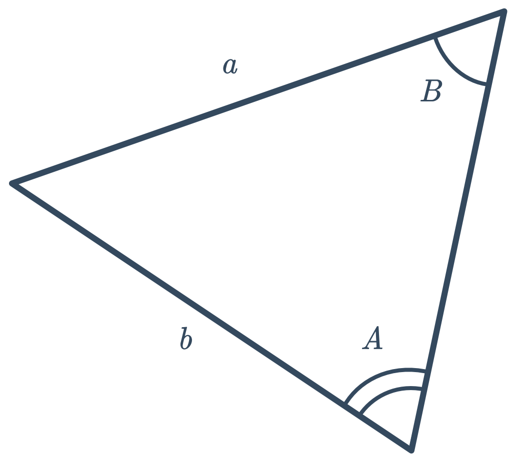


d Two of the sides and an angle included between them are known:

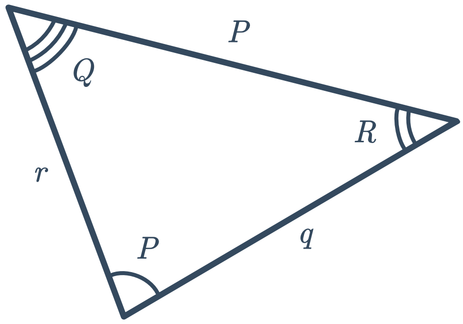


2 For each of the following triangles, write an equation relating the sides and angles using the sine rule:

a



b

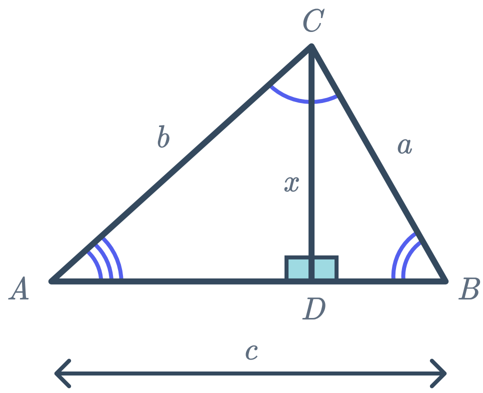


3 Consider the following diagram:

a Find an expression for $\sin A$ in $\triangle ACD$.

b Find an expression for $\sin B$ in $\triangle BDC$. Then make x the subject of the equation.

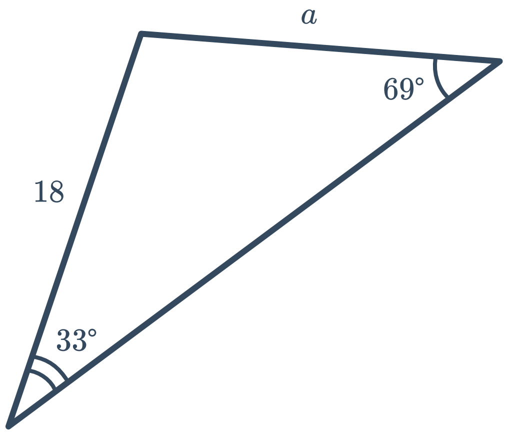
c Substitute your expression for x into the equation from part (a), and rearrange the equation to form the sine rule.



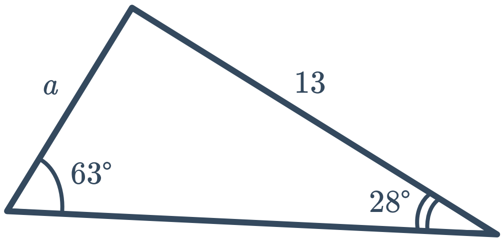
Unknown sides

4 For each of the following triangles, find the side length a using the sine rule. Round your answers to two decimal places.

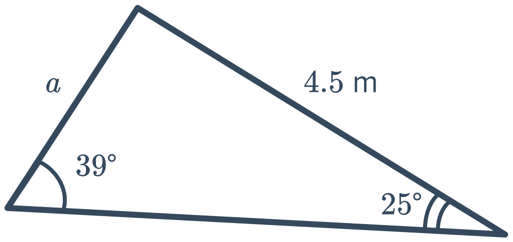
a



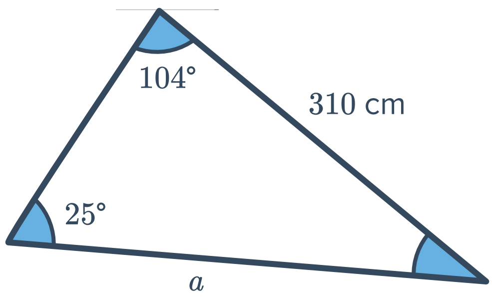
b



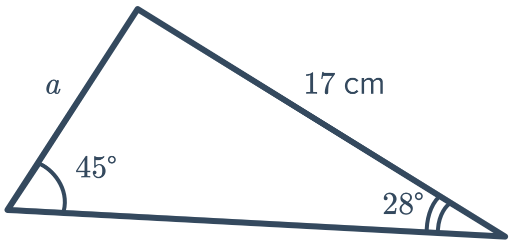
c



d

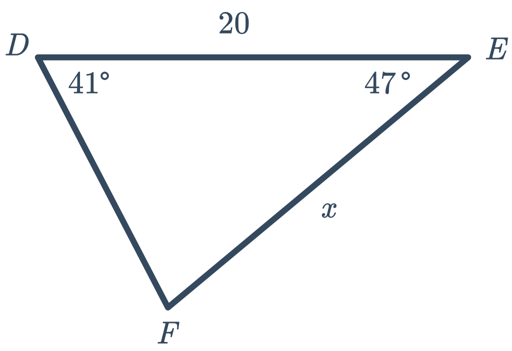


e

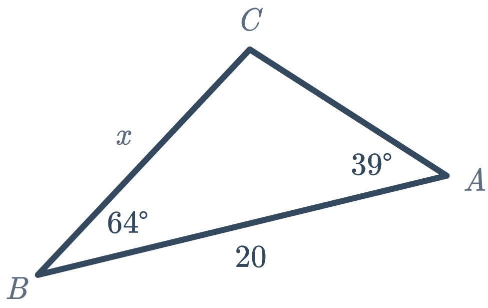


5 For each of the following triangles, find the length of side x, correct to one decimal place:

a



b

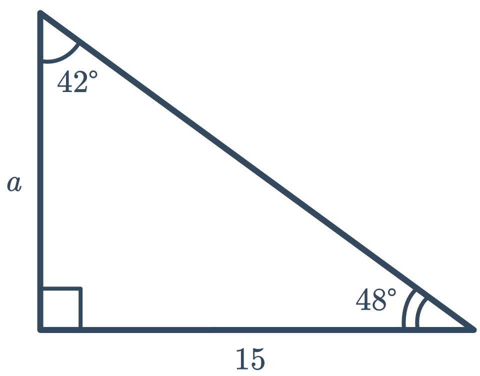


6 For each of the following triangles:

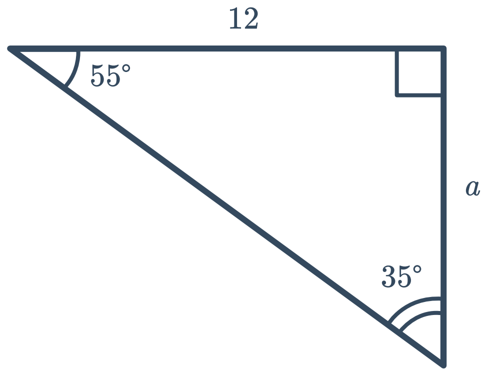
i Find the value of a using the sine rule. Round your answer to two decimal places.

ii Use another trigonometric ratio and the fact that the triangle is right-angled to calculate and confirm the value of a. Round your answer to two decimal places.

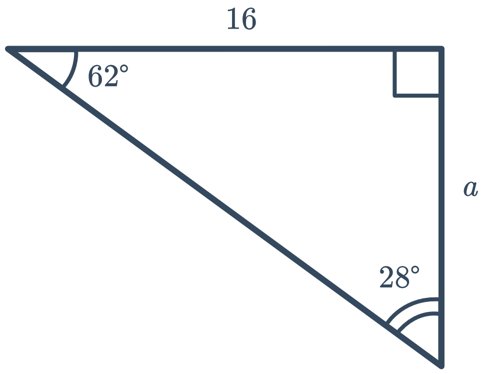
a



b



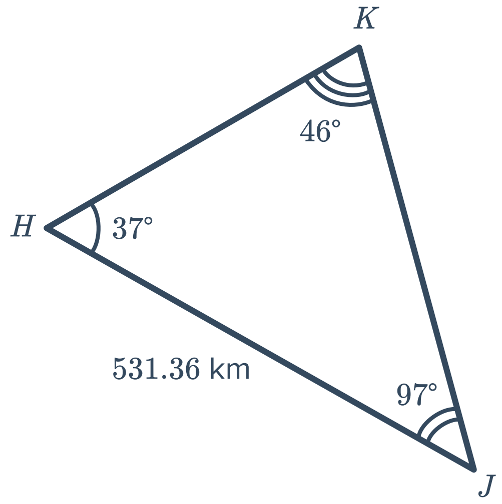
c



7 Consider the following triangle:

a Find the length of side HK to two decimal places.

b Find the length of side KJ to two decimal places.

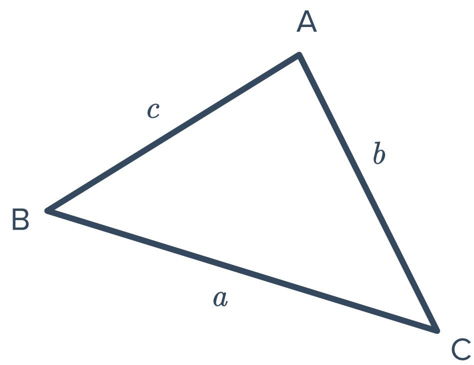


8 Consider the triangle with $\angle C = 72.53^\circ$ and $\angle B = 31.69^\circ$, and one side length $a = 5.816$ m.

a Find $\angle A$. Round your answer to two decimal places.

b Find the length of side b. Round your answer to three decimal places.

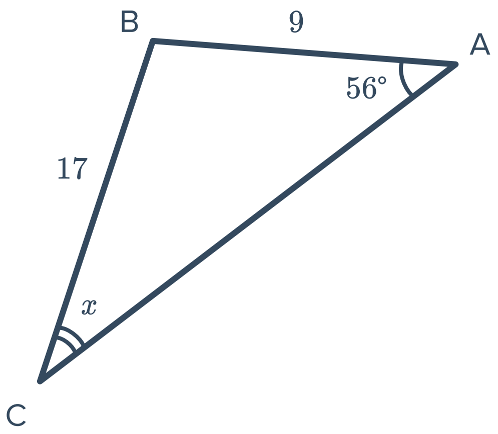
c Find the length of side c. Round your answer to three decimal places.



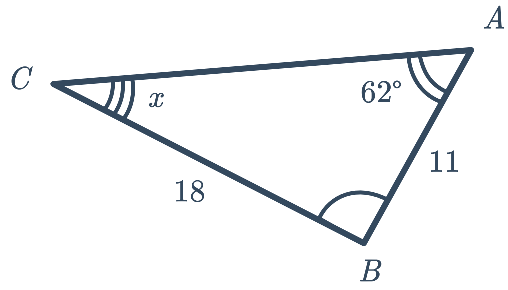
Unknown angles

9 For each of the following diagrams, find the value of the angle x using the sine rule. Round your answers to one decimal place.

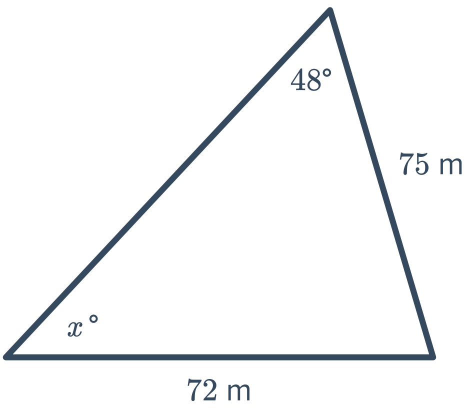
a


B
9
A
56°
17
x
C

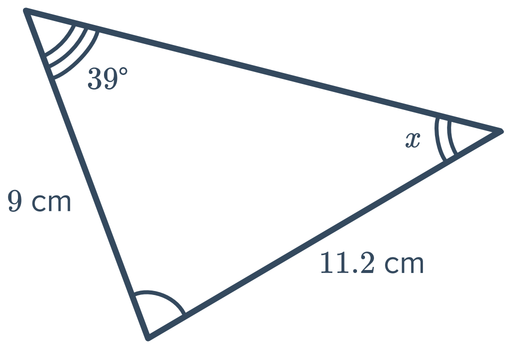
b


A
C
x
62°
11
18
B

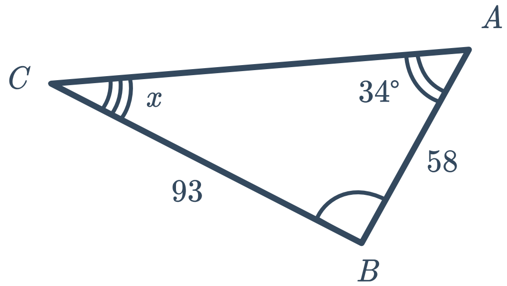
c


48°
75 m
x °
72 m

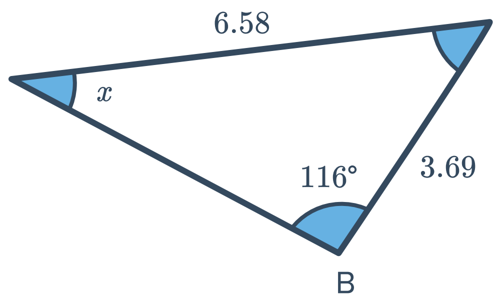
d


39°
x
9 cm
11.2 cm

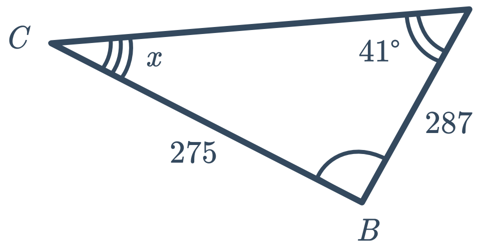
e


A
C
x
34°
58
93
B

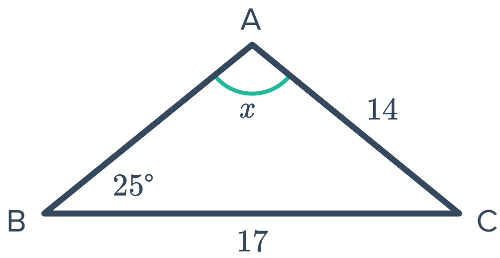
f


6.58
x
116°
3.69
B

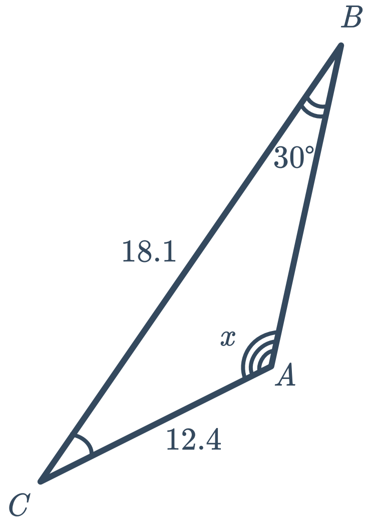
g



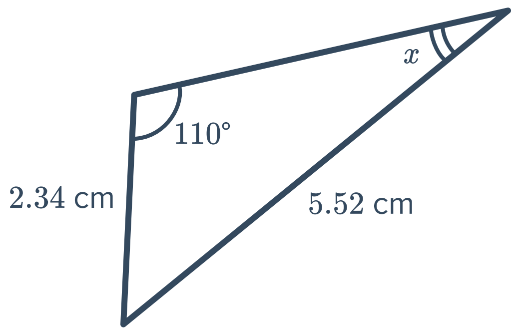
h



i



j



10 For each of the following acute angled triangles, calculate the size of $\angle B$ to the nearest degree:

a $\triangle ABC$ where $\angle A = 57^\circ$ side $a = 156$ cm and side $b = 179$ cm

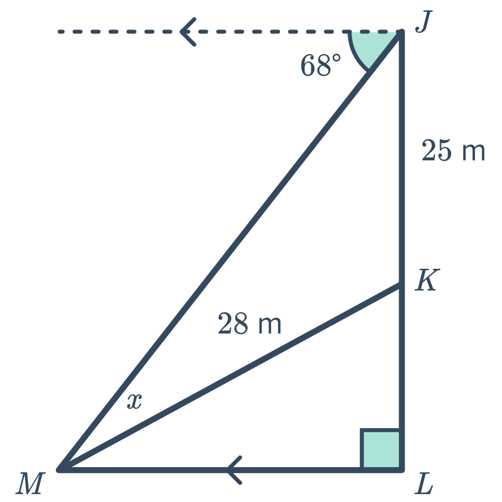
b $\triangle ABC$ where $\angle A = 48^\circ$ side $a = 2.7$ cm and side $b = 1.9$ cm

11 The angle of depression from J to M is 68 The length of JK is 25 m and the length of MK is 28 m as shown:

Find the following, rounding your answers to two decimal places:

a Find x, the size of $\angle JMK$.

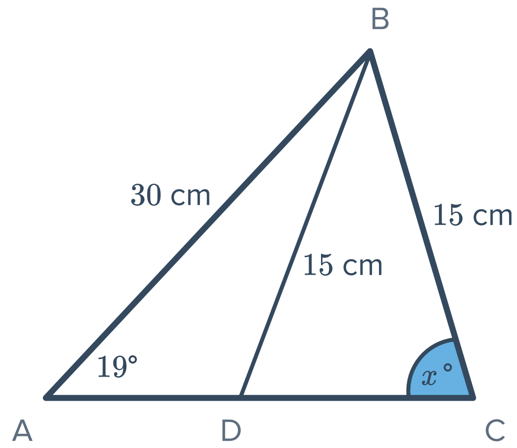
b Find the angle of elevation from M to K.



12 Consider $\triangle ABC$ below:

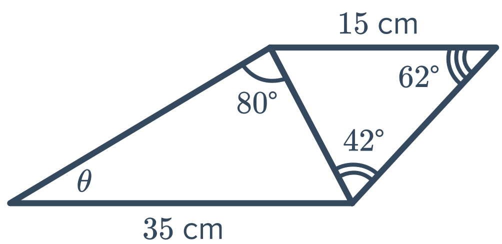
a Find x to the nearest degree.

b Find $\angle ADB$ to the nearest degree.



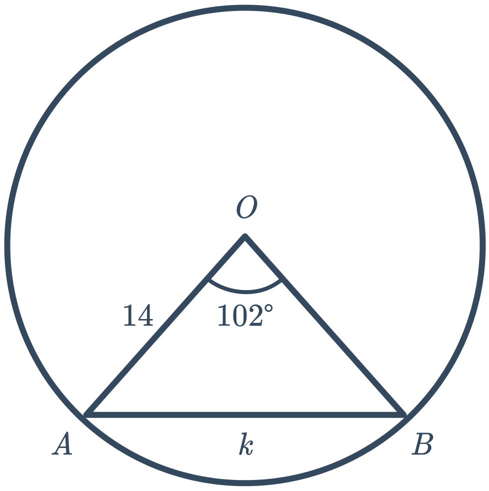
13 Consider the following diagram of a quadrilateral:

Find the value of θ, correct to two decimal places.



Applications

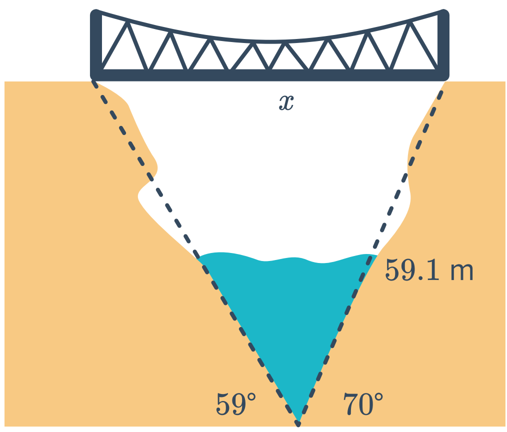
14 Consider the following diagram:



a Find the size of $\angle OBA$.

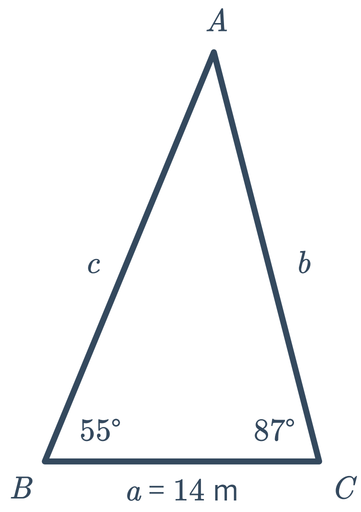
b Find the length of k, to two decimal places.

15 A bridge connects two towns on either side of a gorge, where one side of the gorge is inclined at $59\degree$ and the other side is inclined at $70\degree$. The length of the steeper incline is 59.1 m.



Find x, the length of the bridge. Round your answer correct to one decimal place.

16 During football training, the coach marks out the perimeter of a triangular course that players need to run around. The diagram shows some measurements taken of the course, where side length $a = 14\ \mathrm{m}$:



a Find the size of $\angle A$.

b Find the length of side c, correct to two decimal places.

c Find the length of side b, correct to two decimal places.

d Each player must sprint one lap and then jog one lap around the triangle. This process is to be repeated 3 times by each player.

If Tara can run $280\ \mathrm{m/min}$, and can jog at half the speed she runs, calculate the time this exercise will take her, correct to one decimal place.